

DARSHAN SHASHIKANT PATHAK

+91 9356478034 @ pathak12darshan@gmail.com [linkedin.com/in/pathakdarshan12](https://www.linkedin.com/in/pathakdarshan12) [Portfolio/Blog](#)
Pune,Maharashtra, India github.com/Pathakdarshan12

PROJECTS

DataVelocity: Metadata-Driven Lambda Architecture

<https://github.com/Pathakdarshan12/DataVelocity-metadata-driven-lambda-platform>

Tech Stack: Snowflake, Kafka, Python, AWS S3, Streamlit, Power BI

- Designed a Lambda architecture unifying batch CSV and Kafka streams in the Bronze layer using Snowflake's COPY INTO and Snowpipe, eliminating duplicate logic across entities and reducing the codebase.
- Built metadata-driven quality engine with 4 validation types (mandatory, range, lookup, duplicate) processing 2M+ daily events with 95% pass rate and centralized error logging.
- Engineered hash-based SCD Type 2 using SHA2_HEX, achieving 10x faster change detection with full history tracking.
- Achieved sub-15 second end-to-end latency (Kafka → Gold) using transactional Bronze loads with COMMIT/ROLLBACK and INGEST_RUN_ID idempotency, preventing duplicate processing.
- Developed Streamlit dashboard tracking pipeline health (success rates), validation metrics (error counts), ingestion latency, and entity-level load status for real-time operational visibility.

Isometrics Healthcare: Multi-Tenant SaaS Analytics Platform

<https://github.com/Pathakdarshan12/Isometrics-Healthcare-Multi-Tenant-SaaS-Analytics-Platform>

Tech Stack: dbt, Snowflake, Great Expectations, Python, Streamlit

- Architected medallion data warehouse for 50 hospital tenants processing over 3 million healthcare records with tenant isolation via Row-Level Security policies enforced across 50+ dbt models.
- Implemented incremental materialization strategies (delete and insert for over 3 million encounters, merge for billing transactions) with clustering on hospital_id and date, achieving query performance improvement (from 45 seconds to 12 minutes) and sub-4-hour data freshness with SLA compliance.
- Built over 30 analytical marts calculating quality metrics (such as 30-day readmissions using lag() window functions, medication safety violations, lab turnaround times) and operational KPIs (including bed occupancy rates and cost attribution by tenant) using dbt with ephemeral intermediate models.
- Deployed GitHub Actions CI/CD pipeline running over 250 dbt tests (including RLS validation, tenant isolation, referential integrity) on ephemeral schemas with automatic teardown, reducing Snowflake warehouse costs.
- Developed a Streamlit monitoring dashboard tracking real-time SLA compliance (such as freshness and quality scores), patient-data access logs capturing access events with violation detection, and per-tenant cost attribution using query history analysis.

SKILLS

Data Engineering

Python, SQL, Snowflake, dbt, Databricks, PySpark, PostgreSQL, Kafka

Data Architecture

ETL/ELT Pipelines, Data Modeling, Dimensional Modeling, SCD, CDC

Cloud & DevOps

AWS (S3, Lambda), Azure (ADLS, ADF), Docker, Git, GitHub Actions, CI/CD, Agile

Analytics & Quality

Power BI, Streamlit, Pytest, dbt tests

EXPERIENCE

Associate Data Quality Analyst (Data Engineering – Snowflake/dbt)

[IntegriChain](#)

01/2025 - Present Pune, India

- Developed automated data quality framework for Snowflake production pipelines by designing Python-based validation with 200+ configurable rules, reducing production defects by 40% and eliminating 60% of manual testing overhead.
- Built reconciliation pipelines using SQL and Python to validate data consistency across pipeline stages, comparing row counts, null patterns, and key distributions between source and target tables ; identifying and resolving data anomalies before production deployment.
- Engineered dbt model validation workflows for incremental models processing 50M+ patient enrollment and claims records, implementing schema compatibility checks, regression testing, and data lineage verification to prevent breaking changes to downstream BI dashboards.
- Reconciled Power BI metrics against Snowflake sources using automated SQL validation scripts, resolving calculation inconsistencies in revenue aggregations and ensuring executive reporting accuracy across 20+ financial dashboards.
- Implemented CI/CD testing infrastructure for dbt project deployments, creating regression test suites that validated backward compatibility across 12 schema migrations.

Data Science Engineering Intern

[KodeZera](#)

02/2024 - 08/2024 Pune, India

- Built batch ETL pipeline extracting over 50,000 daily records from PostgreSQL using SQL queries with complex joins and aggregations, applying transformations such as type casting, null handling, and business logic in staging layers, and loading curated datasets into Snowflake analytics tables.
- Implemented incremental data loading by designing timestamp-based watermarking logic that processed only new or modified records since the last run, reducing daily pipeline execution time across source tables.
- Developed data quality validation using SQL-based checks for null values, duplicate records, orphan foreign keys, and range validations across fact and dimension tables, identifying data quality issues and collaborating with the engineering team to fix upstream data sources.
- Optimized Snowflake query performance by analyzing query execution plans, rewriting inefficient joins, and applying clustering keys on date columns for frequently filtered tables, achieving faster query execution for analytical workloads.
- Monitored pipeline runs using execution logs and dashboards, validated record counts after loads, and escalated failures with clear findings during weekly reviews.

EDUCATION

Master of Computer Application

[Pimpri Chinchwad College of Engineering](#)

08/2023 - 04/2025

CGPA
9.23 / 10

Bachelor of Science in Computer Science

[Savitribai Phule Pune University](#)

08/2020 - 04/2023

CGPA
8.92 / 10